

Two cylindric Murnaghan–Nakayama rules are not the same theorem

A dictionary, two located errata, and 16 791 verified coefficients

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24 September 2026

Abstract

Alexandersson and Kantarcı Oğuz [1] prove a Murnaghan–Nakayama (MN) rule for cylindric skew Schur functions and remark of their own main theorem that “a similar formula *seems to appear*” in Korff–Palazzo [2, Eq. (145)]. This note settles the hedge. The answer is *no*: the two formulas are different theorems, and the difference is not a convention. Writing D for a cylindric skew diagram on the cylinder $\mathfrak{C}_{x,y}$ and D' for its conjugate, [1] expand s_D over ribbon data on D , whereas [2] expand s_D over ribbon data on D' ; classically these two statements coincide because $\omega(s_{\lambda/\mu}) = s_{\lambda'/\mu'}$, but cylindrically $\omega(s_D) \neq s_{D'}$ in general (verified: 67 of 357 small shapes), so the coincidence fails and the two rules carry genuinely different multiplicities on fully wound ribbons.

En route, both sources are found to contain a precisely locatable defect, each confined to the genuinely cylindric case and each invisible in the non-winding regime. **(A)** [1] define the height of a stacked ribbon twice and assert the two definitions agree; they differ by ℓ , the number of loop ribbons peeled. Their closed formula $\text{ht}(D) = \ell y + \text{ht}(F)$ gives the wrong sign whenever ℓ is odd. Their own worked example `ex:mnrule` uses the *correct* value (3, not 2) and its printed power-sum expansion is right; the prose formula contradicts the example. **(B)** [2]’s character $\chi_{\bar{\lambda}/d/\bar{\mu}}(\nu)$, as literally defined (a sum over cylindric ribbon plane partitions weighted by a pure sign), omits a multiplicity k carried by the matrix element it is asserted to equal. The minimal counterexample is a single circular ribbon: $n = 4$, $k = 2$, $\bar{\lambda} = \bar{\mu} = (0, 0)$, $d = 1$, $\nu = (4)$, where the definition gives -1 and the truth is -2 .

Every claim below is checked against an independent arbiter: the cylindric Schur function computed directly from its tableau definition. The corrected rules are then shown to be exactly equivalent to Korff [3, Lemma 5.3(i),(ii)], whose $t \rightarrow 1$ limit carries the missing factor $(-1)^k(n - k)$ explicitly; so what is new here is the comparison, the dictionary, and the two errata, not either rule.

1 The question

[1, src l.232] writes, of `thm:cylindricMNRule`:

“In this paper, we give a combinatorial proof of such a formula in `thm:cylindricMNRule`. A similar formula seems to appear in [2, Eq. (145)].”

Nobody has settled whether “similar” means “the same”. Both statements are MN rules; both are finite sums over finite sets; both are therefore decidable on small data. This note decides them.

Remark 1 (honest limit on one citation). [1] cite the published version of [2] (*Algebraic Combinatorics* **3**(1), 2020). I hold only the e-print [arXiv:1804.05647](https://arxiv.org/abs/1804.05647). Counting numbered environments

strictly before and including `\label{cylSchur}` in the e-print gives 140 `equation` plus 5 single-line `eqnarray` environments = 145, an exact hit, and the content agrees ([2] gloss `\eqref{cylSchur}` as “a cylindric or affine version of the Murnaghan–Nakayama rule”). This is strong evidence, *not* proof, that the two renderings number alike. Throughout I cite by label, `\eqref{cylSchur}`, and never write “Eq. (145)” as though I had read that rendering.

2 Conventions, and the dictionary

2.1 The AKO side

Following [1, 1.313–330, 462–472]: $\mathfrak{C}_{x,y}$ is $\mathbb{Z}^2$ with $(0,0)$ identified with $(x,-y)$. In matrix coordinates (r,c) (row index increasing downwards in the drawing) the identification is

$$(r,c) \sim (r+y, c+x), \quad (1)$$

so the cylinder has y rows and x columns per period; in particular its column set is $\mathbb{Z}/x\mathbb{Z}$. A *cylindric diagram* D is the region between two boundary loops; concretely I record it as row intervals $[\text{lo}_r, \text{hi}_r]$, $r \in \mathbb{Z}$, both weakly increasing in r and satisfying $\text{lo}_{r+y} = \text{lo}_r + x$, $\text{hi}_{r+y} = \text{hi}_r + x$. Set $n := x + y$.

A *ribbon* is a connected cylindric diagram with no 2×2 block; a *loop ribbon* is a ribbon with exactly n boxes. The height of a non-loop ribbon is (number of rows it meets) -1 , equivalently its number of vertical edges; all loop ribbons have height y . Peeling loop ribbons off the outer boundary writes a *stacked ribbon* as $R = L_1 \cup \dots \cup L_\ell \cup F$ with F a loop ribbon (*pure*) or a non-loop ribbon (*non-pure*). A *cylindric semistandard Young tableau* (CSSYT) of shape D is a filling whose entries weakly increase as the column index increases and strictly increase as the row index increases (i.e. upwards in the drawing); $s_D = \sum_{T \in \text{CSSYT}(D)} x^T$.

Lemma 2 (well-definedness). *For all 364 stacked ribbons on the 11 cylinders $\mathfrak{C}_{x,y}$ with $x+y \leq 6$ and $|R| \leq 8$, every peeling yields the same triple $(\ell, \text{pure?}, \text{ht}(F) \bmod 2)$. Hence the weight assigned below does not depend on the peeling.*

Lemma 3 (the width is the whole cylinder). *Every loop ribbon occupies all x cylindrical columns. Consequently $\text{wd}(R) = x$ for every pure stacked ribbon R , independently of R .*

Proof. Read a loop ribbon as a cyclic lattice path: traversing one period it takes x steps right and y steps up/down, returning to its translate by (1) . It therefore meets a contiguous block of $x+1$ columns $c_0, \dots, c_0+x$ of the universal cover, whose reductions modulo x exhaust $\mathbb{Z}/x\mathbb{Z}$ (with c_0 and c_0+x the same cylindrical column). A pure stacked ribbon contains a loop ribbon, so it too meets all x . $\square$

2.2 The KP side, and the translation

[2] work with boxed partitions $\bar{\lambda}$ in the $k \times (n-k)$ box, cylindric loops $\{(i, \bar{\lambda}_i)\}$ with $\bar{\lambda}_{i+k} = \bar{\lambda}_i - (n-k)$, and the shape $\bar{\lambda}/d/\bar{\mu}$ whose row i carries the columns $\bar{\mu}_i + 1, \dots, \bar{\lambda}_{i-d} + d$. Their *circular ribbon* has n boxes and height k .

Proposition 4 (dictionary). *Put $x = n - k$ and $y = k$. Then $\bar{\lambda}/d/\bar{\mu}$ is the cylindric diagram on $\mathfrak{C}_{x,y}$ given by*

$$\text{lo}_r = \bar{\mu}_{k-r} + 1, \quad \text{hi}_r = \bar{\lambda}_{k-r-d} + d, \quad r = 0, \dots, k-1, \quad (2)$$

extended by (1). Under this identification KP's circular ribbons are exactly AKO's loop ribbons, and KP's cylindric ribbons are exactly AKO's stacked ribbons. Moreover $|\bar{\lambda}/d/\bar{\mu}| = |\bar{\lambda}| - |\bar{\mu}| + nd$, and transposing the diagram realises $\bar{\lambda} \mapsto \bar{\lambda}'$, $\bar{\mu} \mapsto \bar{\mu}'$ up to a rotation of the cylinder and a 180° rotation of the plane (both of which preserve s_D).

Proof and calibration. The row reversal $i = k - r$ is forced: $\bar{\lambda}$ is weakly decreasing in i while hi must be weakly increasing in r . Periodicity: $r \mapsto r + k$ sends $i \mapsto i - k$ and $\bar{\lambda}_{i-k} = \bar{\lambda}_i + (n - k) = \bar{\lambda}_i + x$, matching (1). The size identity is the computation $\sum_{i=1}^k (\bar{\lambda}_{i-d} + d - \bar{\mu}_i) = |\bar{\lambda}| + d(n - k) + kd - |\bar{\mu}|$. A circular ribbon has $(n - k) + k = x + y = n$ boxes and $k = y$ vertical edges, i.e. is a loop ribbon. *Calibration:* for $d = 0$ the identity $s_{\bar{\lambda}/0/\bar{\mu}} = s_{\bar{\lambda}/\bar{\mu}}$ (the ordinary skew Schur function), asserted by [2] just below `\eqref{cylSchur}`, holds for every case tested, and the whole of (2) is independently confirmed by the $d = 0$ run of Theorem 11 below (3731 coefficients, 0 mismatches). $\square$ $\square$

Remark 5 (the dictionary is where the trap was). It is tempting to read (??) as the assertion $\omega(s_D) = s_{D'}$, since [2] carry both halves of ω explicitly (the factor ϵ_ν and the conjugates $\bar{\lambda}', \bar{\mu}'$). That reading is *wrong*, and it is refuted by the smallest cylindric shape there is: on $\mathfrak{C}_{1,1}$ the two-box diagram $lo = (0)$, $hi = (1)$ is self-conjugate and $s_D = e_2$, while $\omega(e_2) = h_2$. Testing 357 shapes on nine cylinders, $\omega(s_D) = s_{D'}$ fails for 67. It is exactly this failure that makes the two rules inequivalent (§7).

3 The reduction: one number per region

Proposition 6. *Suppose a weight function w on cylindric diagrams satisfies an MN rule*

$$s_D = \sum_{\nu} \frac{p_{\nu}}{z_{\nu}} \sum_{\substack{\emptyset = D_0 \subset D_1 \subset \dots \subset D_{\ell} = D \\ |D_j/D_{j-1}| = \nu_j}} \prod_j w(D_j/D_{j-1}) \quad (3)$$

for all cylindric diagrams D . Then for every cylindric diagram R with $|R| = r$,

$$w(R) = \langle s_R, p_r \rangle.$$

In particular w is unique, and any two candidate MN rules may be compared region by region.

Proof. Apply (3) to $D = R$ and $\nu = (r)$. A chain with a single step of size r is $\emptyset \subset R$ and there is exactly one, so the coefficient of p_r/z_r in s_R is $w(R)$. Since $s_R = \sum_{\nu} c_{\nu} p_{\nu}/z_{\nu}$ and $\langle p_{\nu}, p_{\rho} \rangle = z_{\nu} \delta_{\nu\rho}$, that coefficient is $\langle s_R, p_r \rangle$. $\square$

This is the instrument. The right-hand side is computed from the CSSYT definition alone, so it is an arbiter independent of both papers. As a first check it reproduces [1, `ex:mnrule`] exactly (Example 8 below).

4 Erratum A: the height of a stacked ribbon

[1, 1.1182–1187] give the height of a stacked ribbon twice:

“...by setting the height of a stacked ribbon $D = L_1 \cup L_2 \cup \dots \cup L_{\ell} \cup F$ to be *the sum of the heights of its constituents* (We can equivalently define the height as *the number of vertical edges of an extended ribbon* wrapping around the shape multiple times and cylindrically covers the entire shape...) As all loop ribbons have height y , we have $ht(D) = \ell y + ht(F)$.”

The two definitions are *not* equivalent. An extended ribbon is a path: on top of the vertical edges internal to the constituents it must traverse one edge at each of the ℓ junctions, and those edges are vertical. So the second definition exceeds the first by ℓ , and the closed formula quoted from the first is off by ℓ .

Theorem 7 (the weight function). *For a cylindric diagram R on $\mathfrak{C}_{x,y}$,*

$$\langle s_R, p_{|R|} \rangle = \begin{cases} 0, & R \text{ not a stacked ribbon,} \\ (-1)^{\ell(y+1)+\text{ht}(F)} \cdot 1, & R = L_1 \cup \dots \cup L_\ell \cup F \text{ non-pure,} \\ (-1)^{\ell(y+1)+y} \cdot x, & R \text{ pure.} \end{cases}$$

Equivalently: AKO's `prop:stackedRibbon` and `thm:cylindricMNRrule` are correct verbatim once `ht` is read as the extended-ribbon count $\ell(y+1) + \text{ht}(F)$ rather than $\ell y + \text{ht}(F)$.

Status: verified exhaustively in range, and derived from [3] in §6; not proved here from first principles. Over all 797 regions on the 13 cylinders $\mathfrak{C}_{x,y}$ with $x+y \leq 6$ and $|R| \leq 8$ (of which 59 are pure and 127 have $\ell \geq 1$): 0 mismatches. The negative controls all break, as they must:

weight variant	mismatches / 797
Theorem 7	0
$\text{ht} = \ell y + \text{ht}(F)$ (as stated in [1])	174
Theorem 7 with $\text{wd} \equiv 1$ (drop the width)	44
Theorem 7 with the global sign flipped	547

That the “drop the width” row is nonzero is the negative control certifying that the width factor is load-bearing in the tested range: it is not an artefact of small x .

Feeding Theorem 7 into (3) reproduces the full multi-part rule: over all cylindric diagrams with $x+y \leq 5$, $|D| \leq 6$, all 874 pairs (D, ν) agree, 0 mismatches.

Example 8 ([1]’s own example contradicts [1]’s own formula). Let D be [1, `ex:mnrule`]: on $\mathfrak{C}_{2,2}$, $\text{lo} = (1, 2)$, $\text{hi} = (3, 3)$, so $|D| = 5$ and $n = 4$. Computing s_D from the CSSYT definition and expanding:

$$s_D = -\frac{p_5}{z_5} + 2\frac{p_{41}}{z_{41}} - 2\frac{p_{311}}{z_{311}} + 4\frac{p_{11111}}{z_{11111}},$$

which is *exactly* the expansion printed in [1] — so the example is right, and the arbiter is calibrated. Now D is a non-pure stacked ribbon with $\ell = 1$ and F a single box, $\text{ht}(F) = 0$. The stated formula gives $\text{ht}(D) = \ell y + \text{ht}(F) = 2$, hence coefficient $(-1)^2 = +1$ for p_5/z_5 . The truth is -1 . The corrected formula gives $\ell(y+1) + \text{ht}(F) = 3$, hence -1 — and 3 is precisely the height [1] annotate this very ribbon with (“Non-pure ribbon with height 3”). The prose formula and the worked example disagree, and the example is the correct one.

Remark 9 (why it survived). The discrepancy is $(-1)^\ell$. It is invisible for $\ell = 0$, i.e. for every ribbon that does not wind — which includes the entire classical specialisation, AKO’s cancellation-free staircase corollary as far as it is exercised by non-winding shapes, and four of the five stacked ribbons in their illustrative example list. A parenthetical “we can equivalently define...” is graded by nothing.

5 Erratum B: the missing multiplicity in χ

[2, (??)] define

$$s_{\bar{\lambda}'/d/\bar{\mu}'} = \sum_{\nu \in \mathcal{P}^+} \frac{\chi_{\bar{\lambda}/d/\bar{\mu}}(\nu)}{z_\nu} \epsilon_\nu p_\nu, \quad \epsilon_\nu = (-1)^{|\nu| - \ell(\nu)}, \quad (4)$$

with $\chi_{\bar{\lambda}/d/\bar{\mu}}(\nu) = \sum_{\bar{\pi}} \prod_{i \geq 0} (-1)^{\text{ht}(\bar{\pi}^{-1}(i)) - 1}$ over cylindric ribbon plane partitions of shape $\bar{\lambda}/d/\bar{\mu}$ and weight ν , and prove that (4) equals the cylindric Schur function of the conjugate shape. Their height is the number of rows occupied, with consecutive pieces sharing a row, so for a cylindric ribbon with ℓ windings and tail F

$$\text{ht}^{\text{KP}} = (\text{ht}(F) + 1) + \ell(k - 1), \quad \text{ht}^{\text{KP}} = m(k - 1) + 1 \text{ for } m \text{ circular ribbons and no tail,} \quad (5)$$

which is exactly their recurrence $\chi_{\bar{\lambda}/d/\bar{\mu}}(r) = (-1)^{k-1} \chi_{\bar{\lambda}/(d-1)/\bar{\mu}}(r - n)$ and their Figure caption ($\text{ht} = 2 + 2 - 1 = 3$ for a ribbon of length $5 + 4$).

Lemma 10 (the two signs already agree). *With $y = k$, for every stacked ribbon R ,*

$$(-1)^{\text{ht}^{\text{KP}}(R) - 1} = \begin{cases} (-1)^{\ell(y+1) + \text{ht}(F)}, & R \text{ non-pure,} \\ -(-1)^{\ell(y+1) + y}, & R \text{ pure.} \end{cases}$$

Proof. Non-pure: $\text{ht}^{\text{KP}} - 1 = \text{ht}(F) + \ell(y - 1)$ and $\ell(y + 1) - \ell(y - 1) = 2\ell$ is even. Pure with $m = \ell + 1$ loops: $\text{ht}^{\text{KP}} - 1 = m(y - 1)$ while $\ell(y + 1) + y = m(y + 1) - 1$, and $m(y + 1) - 1 - m(y - 1) = 2m - 1$ is odd. $\square$

So, once Erratum A is applied, *the sign conventions of [1] and [2] reconcile perfectly on non-winding-tail ribbons and differ by exactly one sign on fully wound ones.* All that is left between them is a multiplicity.

Theorem 11 (what (4) needs). *Let $D = \bar{\lambda}/d/\bar{\mu}$ on $\mathfrak{C}_{n-k,k}$ and $E = \bar{\lambda}'/d/\bar{\mu}'$ on $\mathfrak{C}_{k,n-k}$. Define $\tilde{\chi}_D(\nu)$ by (3)-style chains on D with per-step weight*

$$v(R) = (-1)^{\text{ht}^{\text{KP}}(R) - 1} \cdot \begin{cases} k, & R \text{ fully wound (all circular ribbons),} \\ 1, & \text{otherwise.} \end{cases}$$

Then $\tilde{\chi}_D(\nu) = \epsilon_\nu \langle s_E, p_\nu \rangle$, i.e. (4) holds with χ replaced by $\tilde{\chi}$.

Status: verified, not proved. Over all $(\bar{\lambda}, \bar{\mu}, d)$ with $n \leq 7$, $0 \leq d \leq 3$ and $|D| \leq 7$ on 14 pairs (n, k) :

multiplicity on a fully wound step	coefficients tested	mismatches
k (Theorem 11)	16 791	0
1 ([2] as literally defined)	16 791	439
$n - k$	16 791	425
1, restricted to $d = 0$ (calibration)	3 731	0

The last row is the calibration demanded of any such comparison: at $d = 0$ nothing winds, the multiplicity is invisible, and (4) reduces to the classical $s_{\lambda'/\mu'} = \sum_{\nu} \epsilon_\nu \chi_{\lambda/\mu}(\nu) p_\nu / z_\nu$ — which passes. So the dictionary of Proposition 4 is right and the $d \geq 1$ failures are real.

Erratum 12. $\chi_{\bar{\lambda}/d/\bar{\mu}}(\nu)$ as defined in [2] is short by a factor k for each part of ν consumed by a fully wound cylindric ribbon. Minimal instance: $n = 4$, $k = 2$, $\bar{\lambda} = \bar{\mu} = (0, 0)$, $d = 1$, so D is a *single circular ribbon* of length 4 on $\mathfrak{C}_{2,2}$, and $\nu = (4)$. The definition gives one ribbon plane partition of height $\text{ht}^{\text{KP}} = k = 2$, hence $\chi = (-1)^{2-1} = -1$. But $E = D$ here and $\langle s_E, p_4 \rangle = +2$ with $\epsilon_{(4)} = -1$, so (4) requires $\chi = -2$.

Remark 13 (where the factor went). The multiplicity is present in [2]’s *proof* and absent only from the combinatorial definition. Their Lemma is stated for the matrix element $\langle v^{\bar{\lambda}}, P_{\nu}^* v_{\bar{\mu}} \rangle$, and the proof reads “ P_r^* adds *all possible* ribbons of length r to $\bar{\mu}$ ”. Their `lem:CR` asserts a *unique* starting diagonal a only under the hypothesis $r \notin n\mathbb{N}$; when $r \in n\mathbb{N}$ the starting diagonal is not unique, several a give the same shape, and the matrix element counts them. A cylindric ribbon *plane partition*, however, records only shapes — the level $\bar{\pi}^{-1}(i)$ is a region, not a region-with-a-marked-starting-diagonal — so the degeneracy is lost in passing from the matrix element to χ . The hypothesis $r \notin n\mathbb{N}$ in `lem:CR` is where the information was, and it is exactly the case excluded there that carries the multiplicity.

6 Reconciliation with Korff (2019), and a proof of Erratum A

Korff [3, `lem:cylMNRrule`] proves a cylindric MN rule for t -deformed cylindric Hecke characters $\chi_t^{\lambda/d/\mu}$, defined by matrix elements at the *conjugates* λ', μ' and satisfying $\text{ch}_t(\chi_t^{\lambda/d/\emptyset}) = s_{\lambda/d/\emptyset}$ ([3, `cor:cylchi2cylschur`]) — so there the ribbon combinatorics and the Schur function live on the *same* cylinder $\mathfrak{C}_{n-k,k}$. In the $t \rightarrow 1$ limit [3, Remark following `lem:cylMNRrule`] his two clauses read

$$(i) \quad \chi^{\lambda/d/\mu}(\nu_1, \dots, \nu_i + n, \dots, \nu_\ell) = (-1)^{k-1} \chi^{\lambda/d-1/\mu}(\nu), \quad (6)$$

$$(ii) \quad \chi^{\lambda/d/\mu}(\nu, n) = (-1)^k (n - k) \chi^{\lambda/d-1/\mu}(\nu). \quad (7)$$

Corollary 14. *Theorem 7 is exactly equivalent to (6)–(7), and [1]’s stated formula $\text{ht} = \ell y + \text{ht}(F)$ is not.*

Proof. With $y = k$ and $x = n - k$, Theorem 7 gives the weight $(-1)^{\ell(y+1)+\text{ht}(F)}$, so raising ℓ by one — adding one winding, which is (6) — multiplies by $(-1)^{y+1} = (-1)^{k-1}$: clause (i). The $\ell = 0$ pure case, a single loop ribbon, has weight $(-1)^y x = (-1)^k (n - k)$: clause (ii). Under the stated formula $\text{ht} = \ell y + \text{ht}(F)$ the first factor would instead be $(-1)^y = (-1)^k$, contradicting (6). $\square \quad \square$

Since (6) is proved in [3] directly from the operator identity $H_{\nu_i+n} = (-1)^{k-1} q H_{\nu_i}$, Corollary 14 upgrades Erratum A from “verified on 797 regions” to a consequence of a proved lemma in the literature.

Remark 15 (novelty gate). The corrected AKO rule is therefore, in the $t \rightarrow 1$ limit, [3, `lem:cylMNRrule`] (2019) — which predates [1] and which [1] do not cite at this point; they cite [2] instead, and [2] is the *further* of the two relatives. What is new here is neither rule: it is (a) the dictionary of Proposition 4 with its $d = 0$ calibration, (b) the reduction of Proposition 6 that makes the comparison decidable, (c) the two errata with minimal counterexamples, and (d) the verdict below. [1]’s own contribution — a combinatorial proof and a closed, non-recursive form — is untouched by Erratum A, which is a defect in one formula, not in the theorem.

7 Verdict on the hedge

Theorem 16 (the two rules are different theorems). *Write $\chi_D^A(\nu)$ for the summand total of [1, `thm:cylindricMNRrule`] on a cylindric diagram D (with the corrected height), and $\tilde{\chi}_D^K(\nu)$ for that*

of Theorem 11. Then

$$\chi_D^A(\nu) = \langle s_D, p_\nu \rangle, \quad \tilde{\chi}_D^K(\nu) = \epsilon_\nu \langle s_{D'}, p_\nu \rangle,$$

so $\chi^A = \tilde{\chi}^K$ would force $\omega(s_D) = s_{D'}$ for all cylindric D — which is false (67 failures in 357 shapes; the minimal one is $s_D = e_2$ on $\mathfrak{C}_{1,1}$, self-conjugate). Concretely the two rules sum over the same sets of stacked = cylindric ribbons with the same signs (Lemma 10), and differ precisely on fully wound steps, by the factor

$$\frac{v(R)}{w(R)} = -\frac{k}{n-k}.$$

The hedge “a similar formula seems to appear” is thus *correct as a hedge and false as an identification*. The two statements are the two sides of level-rank duality $qH^*(\text{Gr}(k, n)) \cong qH^*(\text{Gr}(n-k, n))$: the multiplicity on a fully wound ribbon is the width of the cylinder on which the *Schur function* lives, $n-k$ for [1] and [3] (ribbons and Schur function on one cylinder), k for [2] (ribbons on one cylinder, Schur function on the conjugate). Classically the two conventions are interchangeable because $\omega(s_{\lambda/\mu}) = s_{\lambda'/\mu'}$; cylindrically ω no longer conjugates the diagram, and the two rules part company exactly where the ribbon winds.

8 Gaps

1. **Theorem 7 and Theorem 11 are verified, not proved from first principles.** Ranges: 797 regions / 874 multi-part pairs / 16 791 KP coefficients, all with 0 mismatches, on $x+y \leq 6$ resp. $n \leq 7$, $|D| \leq 8$, $d \leq 3$. Erratum A is additionally implied by [3, (6)] (Corollary 14); Erratum B is not — it rests on the computation plus the structural reason given in the Remark after Erratum 12.
2. **The extended-ribbon reading of Erratum A is an explanation, not a proof.** I verified that AKO’s two definitions differ by ℓ as a closed formula, and that each junction between consecutive constituents carries at least one vertical edge; but I did *not* prove that an extended ribbon always exists, nor that it must turn vertically at every junction. In 37 of 127 tested stacked ribbons with $\ell \geq 1$ the junction carries *two* vertical adjacencies, of which a path uses one — consistent with, but not a proof of, the count $\ell(y+1) + \text{ht}(F)$.
3. **Equation numbering.** “Eq. (145)” is identified with `\eqref{cylSchur}` by an environment count in the e-print, not by reading the published rendering (see Remark 1.1).
4. **Corollary 14 leans on an implicit extension.** [3, cor:cylchi2cylschur] is stated for the *non-skew* case $\text{cht}(\chi_t^{\lambda/d/\emptyset}) = s_{\lambda/d/\emptyset}$. I have used it in the skew form $\chi^{\lambda/d/\mu}(\nu) = \langle s_{\lambda/d/\mu}, p_\nu \rangle$, which [3] treats as understood but does not display. If that extension fails, Erratum A reverts from “proved” to “verified on 797 regions”, which is still its status independently.
5. **Lemma 2** (uniqueness of the peeling invariants) is verified in range, not proved.

References

- [1] P. Alexandersson, E. Kantarcı Oğuz, *Cylindric Schur functions*, [arXiv:2311.07382](https://arxiv.org/abs/2311.07382); source file `Cylindric-Schur-functions-July-merge.tex`, cited by internal label and source line (thm:cylindricMNRrule l.1269, prop:stackedRibbon l.1240, ex:mnrule l.1284, height discussion l.1182, hedge l.232, def:cylindricSkewSchur l.752).

- [2] C. Korff, D. Palazzo, *Cylindric Hecke characters and Gromov–Witten invariants via the asymmetric six-vertex model*, `arXiv:1804.05647` (published as *Algebr. Comb.* **3** (2020) 191–247); cited by internal label (`\eqref{cylSchur}` l.1955, the matrix element Lemma l.1928 and its proof l.1940–1955, `lem:CR` l.1922, ϵ_ν l.489).
- [3] C. Korff, *(cylindric Hecke characters / asymmetric six-vertex model)*, `arXiv:1906.02565`, draft `korff-1906.02565-draft_02.tex`; `lem:cylMNrule` l.1770, its $t \rightarrow 1$ Remark l.1831, `cor:cylchi2cylschur` l.2037. Cited by internal label and source line, per the same convention.